ATLAS: Anytime-Valid Certification of World-Model Faithfulness by Betting

Anonymous authors
Paper under double-blind review

Abstract

A world model is useful only while its imagined rollouts stay close to what the environment would have done, and that closeness depends on the rollout horizon and changes during deployment. Offline scores such as Fréchet Video Distance or held-out return gaps are computed once and cannot follow either effect. We study how to monitor and certify a deployed world model sequentially, with error guarantees that hold uniformly over time and that remain valid when the planner's own actions, chosen through the model, determine which states the monitor sees. ATLAS bets against the null "horizon h is ε-faithful" with a test martingale for each horizon, and uses the same betting machinery in estimation mode to produce confidence sequences on the faithfulness gap. The first object, the *revocation frontier*, controls false alarms; the second, the *certified horizon*, controls the opposite error, trusting a horizon that has drifted, which is what a planner gating its imagination depth needs. We prove time-uniform validity, a corrected confidence sequence for the running average of conditional means with a simulation-lemma-style consequence for per-horizon total variation, and a finite-sample detection delay bound with a horizon-resolved growth rate. On a simulator with known ground truth the delay law predicts measured delays within 16% across horizons and the certified bound is never violated. On a closed-loop stream where a model-based planner steers the state, CUSUM and a conformal martingale monitoring raw prediction error alarm in every run when the planner heads for high-noise states, and every raw-error monitor reaches false-alarm rates of 0.40–0.73 under a monitor-aware planner, while ATLAS stays at or below 0.013. On three real video shifts, replicated over 30 random splits and compared with standard change detectors on identical streams, ATLAS detects every shift but needs a tolerance margin to keep false alarms near the nominal level. We also report negative results: pooling overlapping horizon chains does not recover the data they discard, and calibrating the tolerance on pooled data voids the conditional guarantee.

1 Introduction

A model-based agent imagines several futures, dozens of steps deep, at every decision, and acts on the one that looks best. That is sensible while the world model matches the environment. Then a joint wears, friction changes, a payload shifts, or the agent's own improving policy carries it into states the model never saw. The imagined futures stay confident, nothing in the monitoring stack fires, because the only number anyone computed was an offline score on held-out data before deployment, and the failure is found by its consequences.

Two properties make the faithfulness of a world model awkward to certify. It *decays with the horizon*: a rollout applies the model recursively, so a small one-step error compounds, and a model that is nearly perfect one step ahead can be useless twenty steps ahead. Faithfulness is therefore a profile over horizons rather than a scalar. It also *drifts over deployment time*, through non-stationarity, the sim-to-real gap, and the feedback loop of model-based reinforcement learning, where the policy generating the data is optimized through the model under test. The data stream is then *endogenous*: it depends on the object being audited.

Current practice scores world models with Fréchet Video Distance (Unterthiner et al., 2018), reconstruction error, or return gaps, all computed once on a fixed set. Conformal prediction (Vovk et al., 2005) and its online variants (Gibbs & Candès, 2021; 2024) are the usual tools for distribution-free uncertainty, but they target marginal coverage of a single prediction. Classical change detectors (CUSUM, Page–Hinkley, windowed tests) control an average run length, so their false-alarm probability grows with monitoring time. What is wanted instead is a running statement, "at round t, rollouts of this model are trustworthy up to horizon h at tolerance ε," whose error probability is controlled uniformly over all rounds, resolved by horizon, indifferent to whether the model predicts pixels or embeddings, and sound when the planner steers the data.

Approach. We build this from the testing-by-betting machinery of anytime-valid inference (Shafer, 2021; Ramdas et al., 2023; Waudby-Smith & Ramdas, 2024). At each round the model emits, for each horizon, an action-conditioned predictive distribution; reality later reveals the outcome. A strictly proper score (Gneiting & Raftery, 2007) turns the pair into a bounded statistic. For each horizon a test martingale bets against the null that the horizon is ε-faithful, which is a *conditional-mean* null, one constraint given the whole past; that is why no exchangeability or independence is needed and why the construction survives policy endogeneity. The same betting machinery, run as a confidence sequence, bounds the faithfulness gap from above. These two outputs answer different questions and control different errors:

- the *revocation frontier* $h^\star(t)$, the largest horizon whose e-processes have not yet crossed $1/\alpha$, is a monitor: a faithful horizon is revoked with probability at most α over all time (an error of availability);
- the *certified horizon* $\hat{h}_\tau(t)$, the largest horizon whose confidence-sequence bound on total variation is below a budget τ, is a certificate: with probability at least $1-\alpha$, no horizon the planner is allowed to use ever has true discrepancy above τ (an error of safety).

A planner that clips its imagination depth to $\min(h^\star, \hat{h}_\tau)$ inherits both guarantees. The two must not be conflated: a bound on false revocations says nothing about trusting an unfaithful horizon.

What is and is not new. Betting e-processes and confidence sequences are not new, and neither is sequential forecast evaluation with e-values (Section 1.1). We do not claim a new inference principle. The contributions are the problem formalization for deployed world models, the analysis of what breaks when the target is multi-step, and a body of evidence about when the construction works and when it does not:

1. **Formalization and multi-step validity** (Sections 2–3). A horizon-indexed, conditional-mean faithfulness null with a tolerance, a statistic whose conditional mean is the object of interest, and an analysis of why consecutive h-step residuals cannot be bet on directly. We validate on non-overlapping chains and show, as a negative result, that the obvious attempt to recover the discarded data by pooling offset chains does not help.
2. **Two dual guarantees** (Section 4): time-uniform false-revocation control; a confidence sequence for the running average of *conditional* means (the constant-mean construction does not certify it); a windowed variant; and a per-horizon total-variation certificate with a simulation-lemma-style value-gap consequence.
3. **A finite-sample delay bound** replacing the asymptotic statement, with a horizon-resolved growth rate that predicts measured delay across horizons.
4. **Evidence** (Section 5): ground-truth validation including the delay law and the certificate; a closed-loop study where the planner steers the stream; a replicated real-video study with competitor monitors on identical streams; and ablations that include the cases where design choices fail.

1.1 Related work

Anytime-valid inference and betting. Ville's inequality (Ville, 1939), game-theoretic probability and e-values (Shafer & Vovk, 2019; Vovk & Wang, 2021; Ramdas et al., 2023), safe testing (Grünwald et al., 2024), and betting confidence sequences (Waudby-Smith & Ramdas, 2024; Howard et al., 2021) supply our tools. Betting

has been applied to two-sample testing (Shekhar & Ramdas, 2024), to auditing fairness of a deployed classifier (Chugg et al., 2023), and to tracking the risk of a deployed model under shift (Podkopaev & Ramdas, 2022).

Sequential forecast evaluation. The closest prior work evaluates probabilistic forecasts sequentially with e-values: calibration tests for probability forecasts (Arnold et al., 2023), comparison of forecasters (Henzi & Ziegel, 2022b; Choe & Ramdas, 2024). These establish that time-uniform inference on forecast scores is possible, and one of them has direct bearing on our setting: a correction to Henzi & Ziegel (2022b) withdraws a supermartingale property for horizons above one, the same overlap phenomenon that we analyze in Section 3.3. They address calibration or relative performance of a forecaster at a fixed lead time. We instead test an absolute tolerance null at every horizon of an action-conditioned simulator, certify the gap from above, and study streams in which the actions depend on the model.

Conformal prediction and change detection. Conformal test martingales (Vovk et al., 2003) monitor exchangeability, and online conformal methods (Gibbs & Candès, 2021; 2024; Tibshirani et al., 2019) relax it for coverage. Change detectors based on cumulative sums, Page–Hinkley, windowing, and Bayesian run-length posteriors (Page, 1954; Bifet & Gavaldà, 2007; Raab et al., 2020; Adams & MacKay, 2007) are tuned to an average run length, and the e-detector framework (Shin et al., 2024) makes the false-alarm guarantee nonparametric. We compare against all of these in Section 5.5.

World models and the simulation lemma. World models range from latent-imagination agents (Ha & Schmidhuber, 2018; Hafner et al., 2023) and joint-embedding predictive architectures (Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025) to interactive and physical-AI simulators (Bruce et al., 2024; NVIDIA et al., 2025). Their evaluation is offline. The simulation lemma (Kearns & Singh, 2002; Janner et al., 2019; Lobel et al., 2024) turns one-step model error into a value gap, with a known one-step error. We replace the known error by a time-uniform certified bound.

2 Problem formulation

2.1 Deployment protocol

A world model M is deployed online with a planner π. At round t the agent is in a (possibly latent) state x_t and the planner chooses actions $a_{t:t+H-1}$ up to a maximum horizon H, using M for imagined rollouts. For each horizon $h \le H$ the model produces an action-conditioned predictive distribution

$$P^M_{t,h} := P^M\big(\,\cdot \mid x_t,\, a_{t:t+h-1}\big) \in \Delta(\mathcal{Y}) \tag{1}$$

over a target space $\mathcal{Y}$ (observations, states, or a learned embedding). Reality reveals the h-step outcome $Y^{(h)}_t$ at time $t{+}h$. Let $\mathcal{F}_k$ be the deployment filtration: everything observable up to the resolution of the k-th prediction of the horizon under consideration, including states, actions, planner internals, and previous outcomes. We assume nothing about independence or exchangeability of the stream. A variable is *predictable* if it is $\mathcal{F}_{k-1}$-measurable.

2.2 Scores, divergence, and the faithfulness null

Fix a strictly proper, negatively oriented score S (Gneiting & Raftery, 2007) and write $d_S(Q,P) = \mathbb{E}_{Y\sim Q}[S(P,Y)] - \mathbb{E}_{Y\sim Q}[S(Q,Y)] \ge 0$ for its divergence, which vanishes iff $P = Q$. For the log score $d_S = \mathrm{KL}$ and Pinsker gives $\mathrm{TV} \le \sqrt{d_S/2}$; for the energy score d_S is a squared energy distance; for a Gaussian predictive the standardized squared error $(y-\mu)^\top \Sigma^{-1}(y-\mu)$ is χ^2_d under calibration, which lets ATLAS score a predicted *embedding* against the realized embedding without decoding to pixels. We assume a modulus: there is a nondecreasing concave ϕ and a probability metric ρ with $\rho(Q,P) \le \phi(d_S(Q,P))$.

Let $Q_{s,h} = \mathrm{Law}(Y^{(h)}_s \mid \mathcal{F}_s)$ be the true conditional law.

Definition 1 (ε-faithfulness)**.** Horizon h is ε_h-faithful if $d_S(Q_{s,h}, P^M_{s,h}) \le \varepsilon_h$ for every issue time s.

We operationalize it through a bounded per-round statistic $Z_k \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$ for each horizon, issued at s_k and resolved at $s_k{+}h$, whose conditional mean $\mu_k := \mathbb{E}[Z_k \mid \mathcal{F}_{k-1}]$ is tied to the divergence in two directions.

Definition 2 (Calibrated and certifiable statistics)**.** The statistic is *calibrated at tolerance* ε_h if $d_S(Q_{s_k,h}, P^M_{s_k,h}) \le \varepsilon_h$ implies $\mu_k \le \varepsilon_h$ (used for the *test*). It is *certifiable with constants* $(c, \bar\rho)$ if $d_S(Q_{s_k,h}, P^M_{s_k,h}) \le c\,\mu_k + \bar\rho$ almost surely (used for the *certificate*). Both are statements about the clipped statistic that is actually computed.

Example 1 (Reference score excess)**.** With a predictable reference R_k and $Z_k = S(P^M_{s_k,h}, Y) - S(R_k, Y)$ we have $\mu_k = d_S(Q, P^M) - d_S(Q, R_k)$. It is calibrated at every ε_h since $d_S(Q, R_k) \ge 0$, and certifiable with $c = 1$ and $\bar\rho \ge \sup d_S(Q, R_k)$, the suboptimality of the reference. A poor reference makes the test conservative and the certificate loose.

Example 2 (Standardized Gaussian excess)**.** For a Gaussian predictive $\mathcal{N}(\mu_M, \Sigma_M)$ take $Z_k = \frac{1}{d}(y - \mu_M)^\top \Sigma_M^{-1}(y - \mu_M) - 1$, clipped to $[-1, z_{\mathrm{hi}}]$. If the true conditional law has the same covariance and differs in mean by m, then $\mu_k = \frac{1}{d} m^\top \Sigma_M^{-1} m$ and $d_S = \mathrm{KL} = \frac{d}{2}\mu_k$, so $c = d/2$, $\bar\rho = 0$, and $\mu_k = 0$ exactly under calibration. The conditional mean is *pivotal*: it does not depend on the state or the action. If the covariance is mismatched, only the one-directional calibration property is retained.

2.3 The endogeneity obstruction

The outcomes $\{Y_t^{(h)}\}$ are neither independent nor exchangeable, and the planner is optimized through M, so the state distribution that the monitor sees is a function of the model. Two things follow. A reference or threshold calibrated offline, on data the planner did not generate, describes a different distribution from the one met in deployment (Section 5.3 measures how much that matters). And a fixed constant tolerance is only meaningful if the statistic's conditional mean does not itself depend on where the planner goes. A conditional-mean null constrains one expectation given the entire past, so it holds for any predictable planner and betting rule; it needs no model of how the planner behaves. What it does not do is guarantee power: a planner may drift to states where the same error is hard to detect, which we quantify in Section 5.3.

3 The ATLAS method

3.1 E-processes and Ville's inequality

An *e-variable* for a null is a nonnegative random variable with expectation at most one under it. A *test supermartingale* is a process $W_0 = 1$, $W_k = \prod_{j \le k} e_j$ with $\mathbb{E}[e_j \mid \mathcal{F}_{j-1}] \le 1$ under the null. The engine of time-uniform validity is classical.

Lemma 1 (Ville 1939)**.** *If* $(W_k)_{k \ge 0}$ *is a nonnegative supermartingale with* $W_0 = 1$ *then* $\mathbb{P}(\sup_k W_k \ge 1/\alpha) \le \alpha$*.*

3.2 The per-horizon betting e-process

Fix a horizon h and a statistic $Z_k \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$ calibrated at ε_h (Definition 2), so that the faithfulness null implies

$$\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \le \varepsilon_h \qquad \text{for all } k. \tag{2}$$

For a predictable bet $\lambda_k \in [0, \lambda_{\max}]$ with $\lambda_{\max} = 1/(z_{\mathrm{hi}} - z_{\mathrm{lo}})$ set

$$e_k := 1 + \lambda_k(Z_k - \varepsilon_h), \qquad W_K^{(h)} := \prod_{k \le K} e_k. \tag{3}$$

Proposition 1 (Per-round validity)**.** *If* (2) *holds and* $\varepsilon_h \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$ *then* $e_k \ge 0$ *and* $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \le 1$*; hence* $W^{(h)}$ *is a test supermartingale.*

Proof. $e_k \ge 1 - \lambda_{\max}(\varepsilon_h - z_{\mathrm{lo}}) \ge 0$, and $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] = 1 + \lambda_k(\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] - \varepsilon_h) \le 1$ because λ_k is predictable. □

The bet governs power, not validity. We use a uniform mixture over a grid $\lambda^{(1)}, \dots, \lambda^{(m)} \in (0, \lambda_{\max}]$: a convex combination of test supermartingales is a test supermartingale, so $W_K^{(h)} = \frac{1}{m}\sum_j \prod_{k \le K}(1 + \lambda^{(j)}(Z_k - \varepsilon_h))$ is

valid, parameter-free, and adapts to an unknown margin at a cost of $\log m$ in the log-wealth. Section 5.6 compares it with a fixed bet and a plug-in bet.

3.3 Multi-step validity: why chains, and why they cannot be pooled

If a horizon-h prediction is formed at *every* round, consecutive residuals share the innovations in a window of length $h-1$, so the residual stream is moving-average of order $h-1$. Conditioning on the past reveals part of the current residual and (2) fails in general even for a faithful model. The same effect underlies the overlap problem in sequential multi-step forecast evaluation (Henzi & Ziegel, 2022a). We restore (2) by running each horizon on *non-overlapping chains* of issue times $c, c+h, c+2h, \ldots$: the k-th chain element is resolved before the next is issued, so $\mathcal{F}_{k-1}$ contains everything the next residual could depend on. The cost is real: horizon h receives a factor h fewer updates.

One might try to win these updates back by running all h offset chains and rejecting when any crosses its threshold (a union bound, costing $\log h$). That is valid but it does not pool evidence: each chain sees only a $1/h$ share of the data and its wealth grows at the same rate per update as the single chain, so taking the first of h crossings buys at most a small order-statistic gain that the $\log h$ in the threshold cancels. Averaging the chains' wealth would pool, but chains live on different filtrations, so the average is an e-value at fixed times and not an e-process (an optional-stopping argument does not go through). Section 5.2 measures the interleaved union: it is valid and no faster than the single chain. We therefore keep the simple estimator and treat the factor h as the intrinsic price of certifying long horizons directly.

3.4 Two outputs: the revocation frontier and the certified horizon

Revocation frontier (monitor). Run one e-process per horizon at level α/H and revoke horizon h at $\tau_h = \inf\{k : W_k^{(h)} \geq H/\alpha\}$, permanently (Ville concerns the event of *ever* crossing). Define

$$h^\star(t) := \max\{h : \text{ no horizon } h' \leq h \text{ is revoked by round } t\}, \tag{4}$$

with $h^\star = 0$ if horizon one is revoked. The prefix rule encodes that a broken short horizon casts doubt on longer ones, and matches the use as a depth clip. The guarantee (Theorem 1) is *no false revocation*: a faithful horizon is lost with probability at most α.

Certified horizon (certificate). In estimation mode the same machinery gives a time-uniform upper bound $U_t^{(h)}$ on the running average of conditional means of horizon h (Lemma 2), hence, for a certifiable statistic, an upper bound on the average divergence and, through the modulus, on total variation: $\widehat{\mathrm{TV}}_h(t) := \min\{1, \phi(cU_t^{(h)} + \bar{\rho})\}$. For a budget τ the certified horizon is

$$\hat{h}_\tau(t) := \max\{h : \ \widehat{\mathrm{TV}}_i(t) \leq \tau \text{ for all } i \leq h\}. \tag{5}$$

The guarantee (Theorem 2) is the opposite error: with probability at least $1-\alpha$, simultaneously over rounds and horizons, the true average discrepancy of every horizon inside $\hat{h}_\tau(t)$ is at most τ. The planner's gate is $\min(h^\star(t), \hat{h}_\tau(t))$. The two do not substitute for each other: $h^\star$ can be high on a model that is mediocre but stable, and $\hat{h}_\tau$ can be low on a faithful model that has not yet accumulated enough evidence.

Change detection. To detect a change in faithfulness rather than test a fixed null we use the Shiryaev–Roberts e-detector (Shin et al., 2024), $R_t = (1 + R_{t-1})e_t$ (with the mixture bet), which accumulates evidence over all candidate changepoints so that wealth lost before the change cannot delay the alarm. Algorithm 1 summarizes the procedure.

4 Theory

Throughout, statistics are bounded, $Z_k \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$ with width $w := z_{\mathrm{hi}} - z_{\mathrm{lo}}$, and bets and references are predictable. No independence or exchangeability is assumed.

Algorithm 1 ATLAS: monitoring and certification of a deployed world model

input: horizons $\{1, \dots, H\}$, tolerances $\{\varepsilon_h\}$, level α, bet grid $\{\lambda^{(j)}\}$, budget τ, constants $(c, \bar{\rho})$
initialize wealth $W^{(h)} \leftarrow 1$, $\texttt{revoked}(h) \leftarrow$ false, and a confidence sequence $\mathrm{CS}^{(h)}$ for each h
for each round $t = 1, 2, \dots$ **do**
  for each horizon h whose chain element resolves at t **do**
    form the bounded statistic Z; update $W^{(h)} \leftarrow \sum_j \frac{1}{m} \Pi_j (1 + \lambda^{(j)}(Z - \varepsilon_h))$ and $\mathrm{CS}^{(h)}$
    if $W^{(h)} \geq H/\alpha$ **then**
      $\texttt{revoked}(h) \leftarrow$ true
    end if
  end for
  $h^\star(t) \leftarrow$ largest prefix with no revoked horizon; $\hat{h}_\tau(t) \leftarrow$ largest prefix with $\widehat{\mathrm{TV}}_i \leq \tau$
  output the gate $\min(h^\star(t), \hat{h}_\tau(t))$
end for

4.1 No false revocation

Theorem 1 (Uniform validity)**.** *Fix $\alpha \in (0, 1)$. If horizon h satisfies (2), then $\mathbb{P}(\exists K : \ W_K^{(h)} \geq 1/\alpha) \leq \alpha$, uniformly over all predictable planners and betting rules. Running the H horizons at level α/H each gives $\mathbb{P}(\exists t, \ \exists h \in \mathcal{H}_0 : \ h$ revoked by round $t) \leq \alpha$ for the set $\mathcal{H}_0$ of faithful horizons.*

Proof. By Proposition 1 and linearity, $W^{(h)}$ is a nonnegative supermartingale with $W_0^{(h)} = 1$ under the null, using only (2) and predictability. Ville's inequality bounds $\mathbb{P}(\sup_K W_K^{(h)} \geq H/\alpha) \leq \alpha/H$ for each horizon, and a union bound over $|\mathcal{H}_0| \leq H$ horizons gives the simultaneous statement. □

The guarantee is conservative: under the null the wealth is a supermartingale that tends to drift down, so empirical rates sit far below α (Section 5.1). It controls false *revocations*; it says nothing about trusting an unfaithful horizon, which is the job of the certificate below.

4.2 A confidence sequence for the average conditional mean

The simulation-lemma-style bound needs the running average $m_K = \frac{1}{K} \sum_{k \leq K} \mu_k$ of the *conditional* means, which vary over the deployment. A betting confidence sequence for a *constant* mean does not certify it: its proof tests $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \geq u$ for every k, which is not implied by an average bound. We use an empirical-Bernstein supermartingale instead.

Lemma 2 (Confidence sequence for the average conditional mean)**.** *Let $\widehat{Z}_k \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$ be predictable, $\psi(\lambda) = -\log(1 - \lambda) - \lambda$, $V_K = \sum_{k \leq K} \big((Z_k - \widehat{Z}_k)/w\big)^2$, and fix a grid $\lambda_1, \dots, \lambda_n \in (0, 1)$ with prior weights ν_j. With probability at least $1 - \alpha$, simultaneously for all $K \geq 1$ and all j,*

$$m_K \ \leq \ U_K := \bar{Z}_K + \frac{w}{\lambda_j K} \Big(\psi(\lambda_j) V_K + \log \frac{1}{\alpha \nu_j} \Big), \tag{6}$$

so the minimum over j is a valid upper confidence sequence. The width is of the empirical-Bernstein order $\sqrt{\hat{\sigma}_K^2 \log(1/\alpha)/K} + w \log(1/\alpha)/K$.

Proof. Rescale to $[0, 1]$. For $y \geq -1$ and $\lambda \in [0, 1)$, $\exp(\lambda y - \psi(\lambda) y^2) \leq 1 + \lambda y$ (Fan et al., 2015). Put $y = \widehat{x}_k - x_k \in [-1, 1]$. Then $\mathbb{E}[\exp(\lambda(\mu_k - x_k) - \psi(\lambda)(x_k - \widehat{x}_k)^2) \mid \mathcal{F}_{k-1}] \leq e^{\lambda(\mu_k - \widehat{x}_k)}(1 + \lambda(\widehat{x}_k - \mu_k)) \leq 1$, for *any* value of the conditional mean μ_k. So $M_K(\lambda) = \exp(\lambda \sum_k (\mu_k - x_k) - \psi(\lambda) V_K)$ is a nonnegative supermartingale, and so is $\sum_j \nu_j M_K(\lambda_j)$. Since $\nu_j M_K(\lambda_j)$ is dominated by the mixture, Ville's inequality gives $\mathbb{P}(\exists K, j : \ \nu_j M_K(\lambda_j) \geq 1/\alpha) \leq \alpha$; rearranging gives (6). □

Corollary 1 (Windows)**.** *Restart the sequence at block starts $s_i = ib$ and give start s_i the level $\alpha/((i+1)(i+2))$. Then with probability at least $1 - \alpha$, simultaneously for all starts and all end times, the average conditional*

mean over $(s_i, t]$ *is at most the corresponding bound. Reading the window with the latest start* $s_i \le t - w_0$ *gives an anytime-valid bound on the* recent *average over a window of length in* $[w_0, w_0 + b)$.

Proof. Apply Lemma 2 at each start with level $\alpha_i = \alpha/((i+1)(i+2))$ and use $\sum_i \alpha_i \le \alpha$. □

The windowed form matters: the full-history average moves slowly after a change, so a certificate meant to follow the present must be read over a window; the price is the larger $\log(1/\alpha_i) \approx 2\log i$ term.

4.3 The certified horizon and a simulation-lemma consequence

Theorem 2 (Anytime-valid certified horizon)**.** *Suppose each horizon's statistic is certifiable with constants* $(c, \bar\rho)$ *(Definition 2) and* $\rho = \mathrm{TV}$ *with modulus* ϕ. *Apply Lemma 2 (or Corollary 1) to the chain of each horizon* $h \le H$ *at level* α/H, *giving bounds* $U_t^{(h)}$ *for the average over a chain index set* $\mathcal{S}_{h,t}$ *(full history or a window). Then with probability at least* $1-\alpha$, *simultaneously for all rounds* t *and horizons* $h \le H$,

$$\frac{1}{|\mathcal{S}_{h,t}|} \sum_{k \in \mathcal{S}_{h,t}} \mathrm{TV}\big(Q_{s_k,h}, P^M_{s_k,h}\big) \;\le\; \min\big\{1,\; \phi\big(cU_t^{(h)} + \bar\rho\big)\big\}. \tag{7}$$

Consequently, with probability at least $1-\alpha$, *for all* t *and all* $h \le \hat h_\tau(t)$ *the left side is at most* τ.

Proof. By certifiability the average divergence over $\mathcal{S}_{h,t}$ is at most $c\,m + \bar\rho$ with m the average conditional mean, which Lemma 2 bounds by $U_t^{(h)}$ for all t, h with probability $1-\alpha$ after a union bound over H horizons. By the modulus, concavity of ϕ (Jensen) and monotonicity, the average of $\mathrm{TV} \le \phi(d_S)$ is at most $\phi(c\,m + \bar\rho) \le \phi(cU_t^{(h)} + \bar\rho)$; total variation is at most one. □

Corollary 2 (Value gap)**.** *Let rewards* $r \in [0, R_{\max}]$ *be a function of the outcome and let actions be held fixed (open-loop), as in the action-conditioned predictive. For discount weights* $w_i \ge 0$, *on the event of Theorem 2 the weighted sum of window-averaged per-step reward gaps satisfies* $\sum_{i \le h} w_i\, G_i(t) \le R_{\max} \sum_{i \le h} w_i \min\{1, \phi(cU_t^{(i)} + \bar\rho)\}$, *where* $G_i(t)$ *is the average of* $|\mathbb{E}_M r(Y^{(i)}) - \mathbb{E}_Q r(Y^{(i)})|$ *over horizon* i*'s chain index set.*

Proof. For fixed actions $|\mathbb{E}_P r - \mathbb{E}_Q r| \le R_{\max}\mathrm{TV}(P, Q)$; apply (7) for each i. □

Remark 1 (Relation to the classical simulation lemma)**.** The classical lemma (Kearns & Singh, 2002) converts a *known* one-step error into an h-step value gap, paying a compounding factor such as $\gamma/(1-\gamma)^2$. Here each horizon's own chain is certified, so there is no propagation step and no concentrability coefficient; the price is that horizon h is certified from a $1/h$ share of the rounds. When the gap profile is constant across issue times in the window, $\sum_i w_i G_i$ is the average discounted return gap. We are not aware of a time-uniform version of this statement in the literature.

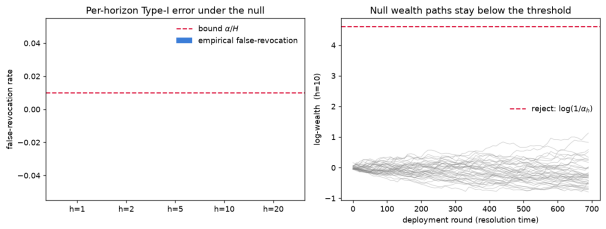
4.4 Detection delay

Take the e-detector R_t built from the mixture bet. For the alarm time $T = \inf\{t : R_t \ge 1/\alpha\}$ the classical argument gives the false-alarm guarantee, and a martingale concentration argument gives a finite-sample delay.

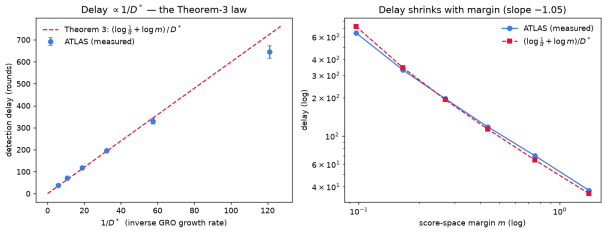
Theorem 3 (Finite-sample detection delay)**.** *Let* $X_k = Z_k - \varepsilon$ *with* $|X_k| \le B'$. (a) *Under the null,* $\mathbb{E}_\infty[T] \ge 1/\alpha$. (b) *Suppose that after a changepoint* τ, $\mathbb{E}[X_k \mid \mathcal{F}_{k-1}] \ge \Delta > 0$ *and* $\mathbb{E}[X_k^2 \mid \mathcal{F}_{k-1}] \le v$ *for all* $k > \tau$, *and that the grid is linear on* $(0, 1/B']$ *with* m *points and* $\Delta \ge 2v/(mB')$. *Let* $\lambda^\star = \min\{\Delta/(2v),\, 1/(2B')\}$. *For every* $\delta \in (0,1)$ *and*

$$n \;\ge\; N := \max\Big\{ \frac{8\log(m/\alpha)}{\lambda^\star \Delta},\; \frac{(80v + 8B'\Delta)\log(1/\delta)}{\Delta^2} \Big\}, \tag{8}$$

we have, for every history, $\mathbb{P}(T - \tau > n \mid \mathcal{F}_\tau, T > \tau) \le \delta$. *Note* $8/(\lambda^\star\Delta) \le 16(v/\Delta^2 + B'/\Delta)$. (c) *For horizon* h *on a non-overlapping chain, with per-chain margin* Δ_h *and second moment* v_h, *the delay in rounds is at most* $h(N_h + 1)$.



(a) Per-horizon false revocation under the null (left) and null wealth paths (right).



(b) Delay against the predicted $(\log \frac{1}{\alpha} + \log m)/D^\star$ (left) and against $1/D^\star$ (right).

Figure 1: Ground-truth validation of Theorems 1 and 3 on the simulator.

The proof is in Appendix A. In the regime where the variance term dominates ($v \gg B'\Delta$), $N \approx 16\, v \log(m/\alpha)/\Delta^2$, which is within a factor $8(1 + \log m/\log(1/\alpha))$ of the Lorden–Lai lower bound $2\sigma^2 \log(1/\alpha)/\Delta^2$ for a Gaussian-like pair with the same mean shift and variance (Lorden, 1971; Lai, 1998). When the bet saturates ($v \lesssim B'\Delta$) the delay scales as B'/Δ instead of $1/\Delta^2$. Part (c) is the horizon-resolved law: horizon h pays a factor h for fewer updates and gains through a larger margin Δ_h as the shift compounds along the rollout. Section 5.2 tests the resulting prediction $h(\log(H/\alpha) + \log m)/D_h^\star$ with $D_h^\star = \sup_\lambda \mathbb{E} \log(1 + \lambda Z)$.

Remark 2 (The e-process alone)**.** For the plain e-process W started at 1, the same argument holds with $\log(m/\alpha)$ replaced by $\log(m/(\alpha W_\tau))$, where W_τ is the wealth at the changepoint: wealth lost before the change delays the alarm. The Shiryaev–Roberts form removes this term because it contains the term started at $\tau+1$ at full wealth.

5 Experiments

The study has four parts. A simulator with known ground truth checks each guarantee against the truth (Section 5.1), including the horizon-resolved delay law (Section 5.2). A closed-loop stream in which a model-based planner steers the state tests the endogeneity claim (Section 5.3). Replicated real-video experiments compare ATLAS with standard monitors on identical streams (Sections 5.4–5.5). Ablations, including the cases where design choices fail, close the section (Section 5.6). Unless stated, $\alpha = 0.05$, the bet is a mixture over 20 values, and horizons are run on non-overlapping chains at level α/H. Code and per-experiment JSON outputs are in the supplementary material.

5.1 Ground-truth validation

The environment is a stable linear-Gaussian latent system whose world model uses the pre-change parameters, so it is exactly faithful until an unmodelled force of size f is added at a known round (Appendix B). The statistic is the standardized Gaussian excess of Example 2.

No false revocation. Over 400 null deployments of 700 rounds, no horizon among $\{1, 2, 5, 10, 20\}$ was ever revoked, against a per-horizon bound of 0.01 and an any-horizon bound of 0.05. In the horizon-resolved study below (seven horizons, 400 null runs of 2,400 rounds) the any-horizon rate is 0.005. The rates sit far below the bound, as expected for an e-process that is a supermartingale under the null.

Detection delay. Sweeping the drift margin over 100 seeds, the measured delay matches $(\log(1/\alpha) + \log m)/D^\star$ to within about 6%, with delay $\cdot\, D^\star \approx 6$, and scales as $1/D^\star$ (Figure 1). In this range the bet saturates, $D^\star \propto$ margin, and delay scales as one over the margin; the $1/\text{margin}^2$ regime of Theorem 3 requires margins below about 0.07.

The frontier and the certified gap. With a drift of size 0.6 opening at round 900 (Figure 2), horizons $1, 2, 3, 5, 8, 12, 20$ are revoked $85, 120, 84, 105, 132, 144, 220$ rounds after the change; the frontier falls from 20

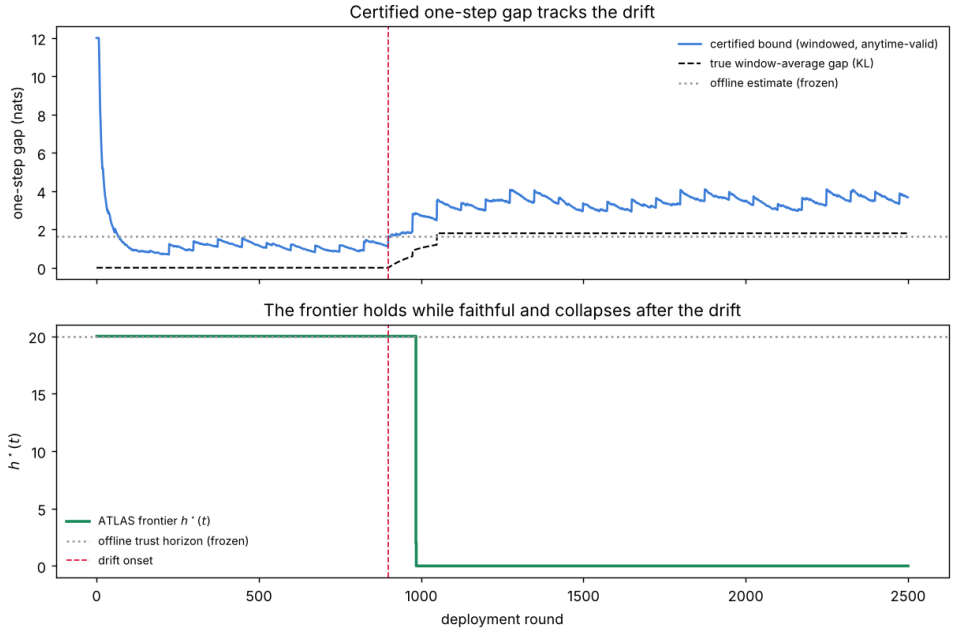


Figure 2: A dynamics drift opens at round 900. Top: the windowed, anytime-valid bound on the one-step gap (solid) against the true window-average gap (dashed) and a frozen offline estimate (dotted). Bottom: the revocation frontier $h^\star(t)$.

Table 1: Certified total-variation bound against the exact value, 150 deployments, drift size 0.25 opening at round 500. Violations: 0 in every cell. Median slack = bound / truth.

	true TV_h	slack +300	slack +800	slack + $\sim$ 2000
$h = 1$	0.307	2.83	2.17	2.29
$h = 3$	0.456	2.96	2.19	2.19
$h = 8$	0.635	2.13	1.58	1.58

to 0, first dropping 84 rounds after the change, while an offline estimate frozen at deployment stays at 20. The windowed certified gap is never below the true window-average gap. It is also loose: 1.63 nats before the change when the true gap is zero and 3.67 after against a true 1.80. A certificate is not a point estimate; it costs data (Section 6).

The certified horizon. Using one chain per horizon with windows of 400 rounds, the certified total-variation bound of Theorem 2 was checked against the closed-form truth over 150 deployments, at checkpoints before the drift and 300, 800 and about 2,000 rounds after it, for $h \in \{1, 3, 8\}$ (Table 1). It was never violated (0/150 at every cell; the guarantee allows 5%) and its slack, bound over truth, is between 1.6 and 3.0 after the change. The slack at $h = 1$ is largest because Pinsker's inequality is loose for a mean shift.

5.2 Horizon-resolved delay and the pooling negative result

Theorem 3(c) predicts that horizon h pays a factor h for fewer updates and gains through a larger margin as the drift compounds. We measure the rounds from the change to revocation for each horizon (150 runs, drift 0.25) and compare with $h(\log(H/\alpha) + \log m)/D_h^\star$, where $D_h^\star = \sup_\lambda \mathbb{E}\log(1 + \lambda Z)$ is estimated by Monte Carlo from the post-change stationary statistic (Table 2, Figure 3).

The prediction is within 16% of the measurement at every horizon and errs on the conservative side. Two features are worth stating plainly. First, in this system $D_h^\star$ grows roughly in proportion to h, so the factor-h

Table 2: Rounds from the drift to revocation, by horizon. Prediction uses only the post-change statistic law. The interleaved estimator takes the first crossing over all h offset chains at the threshold raised by $\log h$.

horizon h	1	2	3	5	8	12	20
growth rate $D_h^\star$	0.016	0.031	0.044	0.067	0.100	0.136	0.186
predicted delay	502	521	543	594	635	699	852
measured, single chain	507	484	497	523	537	585	732
measured, interleaved	507	476	494	554	617	712	923

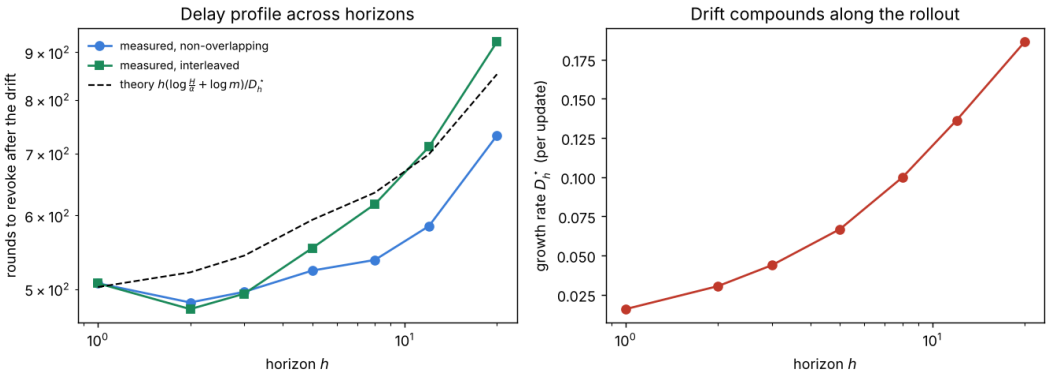


Figure 3: Left: measured delay (single chain and interleaved) against the prediction. Right: the growth rate $D_h^\star$ rises with the horizon because the drift accumulates along the rollout.

loss is almost exactly offset and delay in rounds is nearly flat in h (between 484 and 732); long horizons are not slower *here*, but nothing in the theory says that must hold for other dynamics. Second, the interleaved estimator, which was meant to win back the data discarded by non-overlap, is no faster and is slower at long horizons (923 against 732 rounds at $h = 20$), as the pooling argument of Section 3.3 predicts. Under the null both estimators stay valid (any-horizon false revocation 0.005 and 0.003).

5.3 Policy-endogenous streams

We test the claim that exchangeability- and calibration-based monitors lose validity when the planner steers the state, on a closed-loop latent world model with state-dependent noise in which the world model is exactly faithful, so every alarm under the null is a false alarm (Appendix B). Four planners generate the stream: exogenous exploration (*exo*); a certainty-equivalent controller run through the model toward a low-noise state (*low*) or toward a high-noise state (*high*); and a monitor-aware planner (*adv*) that runs ATLAS's own e-process on the stream it produces and moves to the high-noise region whenever the wealth is positive. The mean noise level induced is 0.24, 0.15, 0.33 and 0.15 respectively. Each monitor sees either the *raw* squared prediction error, the quantity practitioners monitor, or the model-*standardized* statistic that ATLAS uses. Tunable detectors are calibrated to a 5% false-alarm rate at 400 rounds on a separate exogenous calibration deployment, the usual offline protocol, then run for 1,200 rounds under each planner (300 runs each).

Three patterns appear in Table 3. (i) Calibrated offline on exogenous data, raw-error monitors do not carry over: CUSUM and the conformal martingale alarm in every run when the planner heads for the high-noise region, and under the monitor-aware planner every raw monitor reaches 0.40–0.73; under the low-noise planner the same monitors alarm almost never, which would mask a real drift. (ii) On the standardized stream the exchangeability-based conformal martingale is valid for *every* planner. That is not an accident: when the model is exactly faithful the standardized statistic is pivotal, identically distributed whatever the planner does, so the planner cannot hurt a monitor that sees it. The change detectors on that stream still inflate with monitoring time (Page–Hinkley $0.08 \to 0.56$). The honest summary is that conformal monitors are fine on a

Table 3: False-alarm rate on a closed-loop stream with a faithful model: at 400 rounds / by 1,200 rounds, 300 runs per cell. Tunable detectors are calibrated on exogenous data to 5% at 400 rounds. "raw" monitors see squared prediction error, "std" monitors the standardized statistic.

stream	monitor	exo	low	high	adv
raw	CUSUM	.07/.22	.00/.00	**1.00/1.00**	.40/.49
raw	Page–Hinkley	.04/.38	.00/.00	.01/.13	.45/.73
raw	ADWIN	.12/.53	.00/.04	.00/.01	.54/.73
raw	KSWIN	.06/.18	.00/.06	.00/.01	.20/.40
raw	Conformal MG	.09/.10	.00/.00	**1.00/1.00**	.44/.49
std	CUSUM	.09/.28	.07/.23	.09/.25	.08/.26
std	Page–Hinkley	.08/.56	.08/.54	.06/.56	.07/.52
std	ADWIN	.03/.19	.05/.20	.05/.16	.04/.17
std	KSWIN	.02/.08	.03/.08	.03/.06	.02/.08
std	Conformal MG	.03/.03	.03/.03	.01/.01	.02/.02
std	**ATLAS**	**.00/.01**	**.00/.01**	**.00/.01**	**.00/.01**

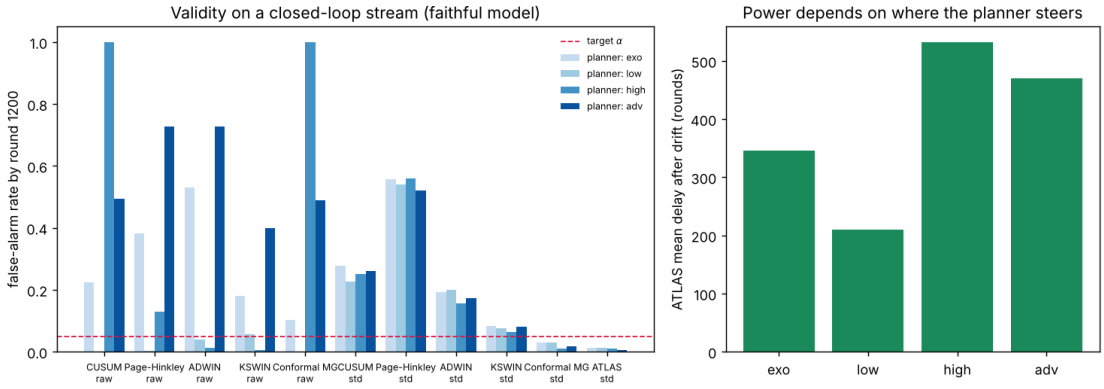


Figure 4: Left: false-alarm rate by round 1,200 for each monitor and planner on the closed-loop stream. Right: ATLAS's mean detection delay after a drift under each planner.

pivotal statistic and ATLAS's advantage is that it needs no pivotality and no tuning, only a conditional-mean bound. (iii) ATLAS never exceeded 0.013 under any planner, including the one that watches its wealth.

Power depends on the planner. After an unmodelled drift of size 0.4 at round 400, ATLAS detected the drift in every run with mean delay 346, 211, 532 and 471 rounds under *exo*, *low*, *high* and *adv*. The planner that seeks high-noise states shields the drift by roughly 1.5× against exogenous exploration, since the same mean shift moves the standardized statistic less where the noise is larger. The conformal martingale on the standardized stream is slower by 1.6–2.2× in every setting and misses the drift in 3–5% of runs. Validity under endogeneity is a theorem; power under endogeneity is an empirical quantity that this study measures for one family of planners and does not bound in general.

5.4 Real video: replicated protocol

Three real shifts are studied with a PCA-plus-linear-dynamics latent world model: Moving MNIST (Srivastava et al., 2015) native speed to double speed, KTH (Schuldt et al., 2004) walking to running, and KTH walking outdoors to indoors. Each of 30 repetitions draws a fresh random split into training, calibration, in-distribution deployment and shifted deployment clips, refits the world model, and sets the tolerance $\varepsilon_h = \bar{z}_h + k\, s_h$ from the calibration clips (mean $\bar{z}_h$ and standard deviation s_h of the clipped statistic, with $k \in \{0, 0.25, 0.5, 1\}$). Competing detectors are tuned on a separate set of 30 repetitions to a 5% false-alarm rate over the in-

Table 4: Replicated real-video study (30 random splits per shift). False alarms over L in-distribution clips and over $3L$; detection rate and delay in clips after the shift; bootstrap 95% intervals in brackets. k is the tolerance margin in standard deviations of the calibration statistic. Tunable detectors are tuned to 5% on separate repetitions.

	FA @ L	FA @ $3L$	detection rate	delay (clips)
Moving MNIST: native → double speed				
ATLAS, $k = 0$	0.17 [0.03,0.30]	0.43 [0.27,0.60]	1.00	4.3 [3.8,4.8]
ATLAS, $k = 0.25$	0.00 [0.00,0.00]	0.00 [0.00,0.00]	1.00	4.6 [4.1,5.1]
ATLAS, $k = 0.5$	0.00 [0.00,0.00]	0.00 [0.00,0.00]	1.00	5.2 [4.6,5.8]
ATLAS, $k = 1$	0.00 [0.00,0.00]	0.00 [0.00,0.00]	1.00	6.6 [5.8,7.5]
CUSUM	0.07 [0.00,0.17]	0.43 [0.27,0.60]	1.00	3.2 [2.7,3.7]
ADWIN	0.07 [0.00,0.17]	0.37 [0.20,0.53]	0.00	—
KSWIN	0.03 [0.00,0.10]	0.27 [0.13,0.43]	0.00	—
Conformal martingale	0.53 [0.37,0.70]	0.67 [0.50,0.83]	1.00	0.3 [0.1,0.5]
KTH: walking → running				
ATLAS, $k = 0$	0.43 [0.27,0.60]	0.47 [0.30,0.63]	1.00	0.3 [0.1,0.5]
ATLAS, $k = 0.25$	0.17 [0.03,0.30]	0.37 [0.20,0.53]	1.00	0.5 [0.2,0.9]
ATLAS, $k = 0.5$	0.07 [0.00,0.17]	0.13 [0.03,0.27]	1.00	1.3 [0.5,2.2]
ATLAS, $k = 1$	0.03 [0.00,0.10]	0.03 [0.00,0.10]	0.70	5.3 [1.7,9.7]
CUSUM	0.97 [0.90,1.00]	0.93 [0.83,1.00]	1.00	0.1 [0.0,0.3]
ADWIN	0.17 [0.03,0.30]	0.43 [0.27,0.60]	0.00	—
KSWIN	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00	—
Conformal martingale	0.70 [0.53,0.87]	0.63 [0.47,0.80]	1.00	0.9 [0.5,1.4]
KTH: outdoor → indoor walking				
ATLAS, $k = 0$	0.33 [0.17,0.50]	0.50 [0.33,0.67]	1.00	1.9 [1.0,3.1]
ATLAS, $k = 0.25$	0.07 [0.00,0.17]	0.23 [0.10,0.40]	0.93	3.1 [1.6,5.2]
ATLAS, $k = 0.5$	0.07 [0.00,0.17]	0.03 [0.00,0.10]	0.77	4.1 [1.7,6.8]
ATLAS, $k = 1$	0.03 [0.00,0.10]	0.03 [0.00,0.10]	0.33	2.3 [0.6,4.5]
CUSUM	0.57 [0.40,0.73]	0.63 [0.47,0.80]	1.00	1.4 [0.9,2.0]
ADWIN	0.10 [0.00,0.20]	0.37 [0.20,0.53]	1.00	6.2 [5.2,7.2]
KSWIN	0.00 [0.00,0.00]	0.00 [0.00,0.00]	1.00	5.6 [4.2,7.2]
Conformal martingale	0.43 [0.27,0.60]	0.57 [0.40,0.73]	1.00	0.4 [0.2,0.6]

distribution deployment length L. We report false alarms over L clips and over $3L$ (in-distribution clips resampled with replacement, standing in for longer monitoring), and detection rate and delay in clips after the shift, with bootstrap 95% intervals.

The picture is mixed, and we report it as such. With the tolerance set at the calibration mean ($k = 0$), ATLAS's false-alarm rate over $3L$ clips is 0.43 on Moving MNIST, 0.47 on KTH walking→running, and 0.50 on KTH outdoor→indoor, all above α: pooled-mean calibration does not satisfy the conditional null (Section 5.6, ablation D). Adding a margin repairs this at a cost in power. On Moving MNIST $k \geq 0.25$ gives no false alarm in 30 repetitions and detection in every repetition, with a delay of 4.6–6.6 clips. On KTH, subject-to-subject heterogeneity makes the statistic far less stable and calibration clips are few: $k = 0.5$ still gives 0.13 false alarms over $3L$ clips for walking→running, and $k = 1$ brings it to 0.03 while detection drops to 0.70; on the subtler outdoor→indoor shift $k = 0.5$ gives 0.03 false alarms and detects in 0.77 of repetitions. The false-alarm rates for $k \geq 0.5$–1 are close to α but nothing here is a guarantee, because the tolerance is estimated. The competing monitors fare worse on false alarms on these real streams: the conformal martingale alarms in 0.43–0.70 of in-distribution repetitions over L clips (exchangeability fails across the temporal and subject structure of the stream), CUSUM in 0.97 on KTH walking→running, and ADWIN and KSWIN, tuned to 5% on separate repetitions, either miss the shift entirely (Moving MNIST, KTH walking→running) or detect it in 5–6 clips (domain shift). Tuning a sensitivity to a target false-alarm rate on a limited number of streams is itself unstable, which is the practical case for a method whose false-alarm control does not rely

Table 5: Simulator, $\alpha = 0.05$: false-alarm rate at the calibration horizon $T_{\text{cal}} = 400$ and at $3T_{\text{cal}}$, and mean detection delay (rounds). BOCPD tuned to the false-alarm target does not detect this drift.

	monitor	FA @ T_{cal}	FA @ $3T_{\text{cal}}$	delay
anytime-valid	**ATLAS**	**0.000**	**0.003**	53.8
	Conformal martingale	0.023	0.023	77.9
tuned	CUSUM	0.113	0.273	12.2
	Page–Hinkley	0.050	0.527	30.4
	ADWIN	0.040	0.140	23.8
	KSWIN	0.040	0.087	33.4
	BOCPD	0.083	0.083	—
	e-detector	0.013	0.197	32.8

on it. We read the table as evidence that the construction is usable on real video when the tolerance is set with a margin and validated, not as evidence that false-alarm control transfers automatically.

5.5 Comparison with competitor monitors on the simulator

On the ground-truth simulator we also compare ATLAS with CUSUM and Page–Hinkley (Page, 1954), ADWIN (Bifet & Gavaldà, 2007), KSWIN (Raab et al., 2020), Bayesian online changepoint detection (Adams & MacKay, 2007), the e-detector (Shin et al., 2024), and a conformal test martingale (Vovk et al., 2003), all on the same one-step statistic stream, with tunable detectors calibrated to 5% at 400 rounds (Table 5). The anytime-valid tests keep false alarms below α at three times the calibration horizon, while every tuned detector inflates (Page–Hinkley from 0.05 to 0.53); the tuned detectors' shorter delay comes from alarming more readily. The conformal martingale is also valid but slower (77.9 against 53.8 rounds) and, as Section 5.3 shows, loses validity when fed non-pivotal scores on a steered stream.

5.6 Ablations and negative results

Table 6 collects five ablations at horizon one (400 null and 200 alternative runs per cell; drift 0.25 at round 500 unless noted).

The mixture is close to the best fixed bet without knowing it (A): a bet that is too small is slow and a large fixed bet is slightly faster but is the one that depends on guessing the margin. Tolerance (B) behaves as designed: when ε is at or above the true gap the revocation rate is at most α; when ε is below it the model really is outside tolerance and revocation grows, slowly, because the gap is tiny. A ramp (C) is detected late, and a short transient is mostly missed: a cumulative e-process is not a detector of brief excursions, and users who care about transients should use a windowed variant.

Ablation D is the most important caution. The log-score advantage over a climatology reference has a *state-dependent* conditional mean, because the reference ignores the state. Setting ε to its pooled null mean makes the null hold on average but not conditionally, and validity fails (0.253 false revocations against a 0.05 guarantee). A constant ε that does hold conditionally (a conservative bound) is so large that the drift is never detected. The standardized Gaussian excess has a pivotal conditional mean and avoids the dilemma. The same issue applies to real video: the tolerances calibrated on pooled calibration clips in Section 5.4 are only empirically safe, which is why the margin k matters there. With heavy-tailed noise (E) the standardized statistic stays valid, since only the covariance enters its conditional mean, and detection is slower.

6 Discussion

What the evidence supports. Time-uniform false-revocation control held in every setting we tried, including a planner that watches the monitor and a stream whose state distribution is a function of the model. The finite-sample delay theory predicts measured delay across horizons to within 16% on a system where the post-change statistic is known. The certified total-variation bound was never violated and was off

Table 6: Ablations. FA = false-revocation rate under the null (or, in B, revocation rate for a model whose true gap is $g = 0.0138$); detect = fraction detected after the drift; delay in rounds.

	variant	FA	detect	delay
A. bet	mixture (20 values)	.025	1.00	392
	plug-in (GRAPA)	.015	1.00	429
	fixed $\lambda = 0.1\lambda_{\max}$	.000	.97	1397
	fixed $\lambda = 0.5\lambda_{\max}$	.030	1.00	401
	fixed $\lambda = \lambda_{\max}$	.038	.99	362
B. tolerance	$\varepsilon = 0$	.125	—	—
	$\varepsilon = 0.5g$	.055	—	—
	$\varepsilon = g$	.025	—	—
	$\varepsilon = 1.5g$	.020	—	—
	$\varepsilon = 2.5g$	.005	—	—
C. drift shape	step	.030	.99	365
	ramp over 300 rounds	.030	.99	607
	transient pulse (150 rounds)	.030	.17	326
D. statistic	standardized excess	.025	1.00	392
	log-score vs. climatology, $\varepsilon = -1$	.000	.00	—
	log-score vs. climatology, $\varepsilon =$ pooled null mean	**.253**	.00	—
E. noise tails	Gaussian	.025	1.00	392
	Student-t_5 (variance-matched)	.007	.98	821

by a factor of 1.6 to 3 after a change. On real video the construction detects the shifts we tried, and what decides whether it is also quiet beforehand is the tolerance.

Limitations.

- *The tolerance is a modeling choice and the dangerous one.* The guarantee needs $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \le \varepsilon_h$ conditionally. A constant tolerance set to the pooled in-distribution mean of a statistic whose conditional mean varies with the state does not satisfy it (ablation D: 0.253 false revocations; real video with $k = 0$: Section 5.4). The standardized Gaussian excess is pivotal and has no such problem, but only for models that output a calibrated predictive covariance. On real video we calibrate ε_h from held-out clips with a margin; that is empirically safe in our replications and not a guarantee.
- *Certificates are expensive.* Pinsker's inequality squares twice on the way from the mean of the statistic to total variation, so certifying a budget τ needs on the order of τ^{-4} rounds at one horizon and h times more at horizon h. In the simulator, windows of a few hundred rounds certify a gap of the size of our drifts but not a small budget on a faithful model (certified 1.63 nats against a true 0 in Figure 2). The certified horizon is therefore informative after large changes and conservative otherwise.
- *Long horizons are intrinsically costly.* Non-overlapping chains give horizon h a $1/h$ share of the rounds; we found no valid way to pool the discarded rounds (Section 3.3). In our simulator a growing margin offsets this; in general it need not.
- *Power is not guaranteed under steering.* A planner heading for states where the same error is harder to see delays detection by about 1.5× in our study; we do not bound this in general.
- *Transients and ramps.* A cumulative e-process detects sustained changes; a 150-round pulse was detected in 17% of runs and a ramp is detected late.
- *Scope of the real-video evidence.* The replicated real-video study uses a lightweight PCA-plus-linear world model on three shifts. The experiments on learned neural world models (convolutional autoencoder, JEPA-style) on KITTI driving and UZH-FPV drone video were single runs that illustrate the interface and

are not replicated; they appear in the supplement of the repository and are not used in any claim above. The benchmark shifts (speed, gait, domain) are large; subtle drifts are the harder and more useful case.

- *Planner effects are not measured.* We certify the model; we do not show that clipping imagination depth to $\min(h^\star, \hat{h}_\tau)$ improves task return, which needs a closed-loop control study.

Practical guidance. Use a statistic whose conditional mean is pivotal if the model provides a calibrated predictive covariance; otherwise add a margin to ε_h and validate it by replaying held-out in-distribution streams for longer than the deployment horizon. Read the certified horizon over a window rather than the full history. Use the revocation frontier as the alarm and the certified horizon as the gate; do not treat the first as evidence for the second.

Future work. Valid pooling of overlapping horizon chains; tolerance learned from a downstream control budget; windowed e-detectors for transients; replicated neural-world-model studies on driving and aerial video; and a closed-loop study of whether the gate improves control performance.

Reproducibility. Code, seeds, and the JSON outputs of every table are in the supplementary material. No new data were collected; the real-video experiments use the public Moving MNIST and KTH datasets.

References

Ryan Prescott Adams and David J. C. MacKay. Bayesian online changepoint detection. *arXiv preprint arXiv:0710.3742*, 2007. doi: 10.48550/arXiv.0710.3742.

Sebastian Arnold, Alexander Henzi, and Johanna F. Ziegel. Sequentially valid tests for forecast calibration. *The Annals of Applied Statistics*, 17(3):1909–1935, 2023. doi: 10.1214/22-AOAS1697.

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture (i-jepa). In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. doi: 10.48550/arXiv.2301.08243.

Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, Koustuv Sinha, Artem Zholus, Sergio Arnaud, Abha Gejji, Ada Martin, Francois Robert Hogan, Daniel Dugas, Piotr Bojanowski, Vasil Khalidov, Patrick Labatut, Francisco Massa, Marc Szafraniec, Kapil Krishnakumar, Yong Li, Xiaodong Ma, Sarath Chandar, Franziska Meier, Yann LeCun, Michael Rabbat, and Nicolas Ballas. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025. doi: 10.48550/arXiv.2506.09985.

Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video (v-jepa). *arXiv preprint arXiv:2404.08471*, 2024. doi: 10.48550/arXiv.2404.08471.

Albert Bifet and Ricard Gavaldà. Learning from time-changing data with adaptive windowing. In *SIAM International Conference on Data Mining (SDM)*, 2007. doi: 10.1137/1.9781611972771.42.

Jake Bruce, Michael Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, Yusuf Aytar, Sarah Bechtle, Feryal Behbahani, Stephanie Chan, Nicolas Heess, Lucy Gonzalez, Simon Osindero, Sherjil Ozair, Scott Reed, Jingwei Zhang, Konrad Zolna, Jeff Clune, Nando de Freitas, Satinder Singh, and Tim Rocktäschel. Genie: Generative interactive environments. In *International Conference on Machine Learning (ICML)*, 2024. doi: 10.48550/arXiv.2402.15391.

Yo Joong Choe and Aaditya Ramdas. Comparing sequential forecasters. *Operations Research*, 72(4):1368–1387, 2024. doi: 10.1287/opre.2021.0792.

Ben Chugg, Santiago Cortes-Gomez, Bryan Wilder, and Aaditya Ramdas. Auditing fairness by betting. In *Advances in Neural Information Processing Systems*, 2023.

Xiequan Fan, Ion Grama, and Quansheng Liu. Exponential inequalities for martingales and applications. *Electronic Journal of Probability*, 20:1–22, 2015.

David A. Freedman. On tail probabilities for martingales. *The Annals of Probability*, 3(1):100–118, 1975.

Isaac Gibbs and Emmanuel Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. doi: 10.48550/arXiv.2106.00170.

Isaac Gibbs and Emmanuel Candès. Conformal inference for online prediction with arbitrary distribution shifts. *Journal of Machine Learning Research*, 2024. doi: 10.48550/arXiv.2208.08401.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.

Peter Grünwald, Rianne de Heide, and Wouter M. Koolen. Safe testing. *Journal of the Royal Statistical Society: Series B*, 86(5):1091–1128, 2024. doi: 10.1093/jrsssb/qkae011.

David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2018. doi: 10.48550/arXiv.1803.10122.

Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*, 2023. doi: 10.48550/arXiv.2301.04104.

Alexander Henzi and Johanna F. Ziegel. Correction to: Valid sequential inference on probability forecast performance. *Biometrika*, 109(4):1181–1182, 2022a. doi: 10.1093/biomet/asac043.

Alexander Henzi and Johanna F. Ziegel. Valid sequential inference on probability forecast performance. *Biometrika*, 109(3):647–663, 2022b. doi: 10.1093/biomet/asab047.

Steven R. Howard, Aaditya Ramdas, Jon McAuliffe, and Jasjeet Sekhon. Time-uniform, nonparametric, nonasymptotic confidence sequences. *The Annals of Statistics*, 49(2):1055–1080, 2021. doi: 10.1214/20-AOS1991.

Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. doi: 10.48550/arXiv.1906.08253.

Michael Kearns and Satinder Singh. Near-optimal reinforcement learning in polynomial time. *Machine Learning*, 49:209–232, 2002. doi: 10.1023/A:1017984413808.

Tze Leung Lai. Information bounds and quick detection of parameter changes in stochastic systems. *IEEE Transactions on Information Theory*, 44(7):2917–2929, 1998. doi: 10.1109/18.737522.

Sam Lobel, Omer Gottesman, and George Konidaris. An optimal tightness bound for the simulation lemma. In *Reinforcement Learning Conference (RLC)*, 2024. doi: 10.48550/arXiv.2406.16249.

Gary Lorden. Procedures for reacting to a change in distribution. *The Annals of Mathematical Statistics*, 42 (6):1897–1908, 1971. doi: 10.1214/aoms/1177693055.

NVIDIA, Niket Agarwal, Arslan Ali, Maciej Bala, Yogesh Balaji, et al. Cosmos world foundation model platform for physical AI. *arXiv preprint arXiv:2501.03575*, 2025. doi: 10.48550/arXiv.2501.03575.

E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954. doi: 10.2307/2333009.

Aleksandr Podkopaev and Aaditya Ramdas. Tracking the risk of a deployed model and detecting harmful distribution shifts. In *International Conference on Learning Representations*, 2022.

Christoph Raab, Moritz Heusinger, and Frank-Michael Schleif. Reactive soft prototype computing for concept drift streams. *Neurocomputing*, 416:340–351, 2020. doi: 10.1016/j.neucom.2019.11.111.

Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023. doi: 10.1214/23-STS894.

Christian Schuldt, Ivan Laptev, and Barbara Caputo. Recognizing human actions: A local SVM approach. In *International Conference on Pattern Recognition (ICPR)*, 2004. doi: 10.1109/ICPR.2004.1334462.

Glenn Shafer. Testing by betting: A strategy for statistical and scientific communication. *Journal of the Royal Statistical Society: Series A*, 184(2):407–431, 2021. doi: 10.1111/rssa.12647.

Glenn Shafer and Vladimir Vovk. *Game-Theoretic Foundations for Probability and Finance*. Wiley, 2019. doi: 10.1002/9781118548035.

Shubhanshu Shekhar and Aaditya Ramdas. Nonparametric two-sample testing by betting. *IEEE Transactions on Information Theory*, 2024.

Jaehyeok Shin, Aaditya Ramdas, and Alessandro Rinaldo. E-detectors: A nonparametric framework for sequential change detection. *The New England Journal of Statistics in Data Science*, 2024. doi: 10.48550/arXiv.2203.03532.

Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov. Unsupervised learning of video representations using LSTMs. In *International Conference on Machine Learning (ICML)*, 2015. doi: 10.48550/arXiv.1502.04681.

Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. doi: 10.48550/arXiv.1904.06019.

Thomas Unterthiner, Sjoerd van Steenkiste, Karol Kurach, Raphaël Marinier, Marcin Michalski, and Sylvain Gelly. Towards accurate generative models of video: A new metric and challenges. *arXiv preprint arXiv:1812.01717*, 2018. doi: 10.48550/arXiv.1812.01717.

Jean Ville. Étude critique de la notion de collectif. *Gauthier-Villars, Paris*, 1939.

Vladimir Vovk and Ruodu Wang. E-values: Calibration, combination and applications. *The Annals of Statistics*, 49(3):1736–1754, 2021. doi: 10.1214/20-AOS2020.

Vladimir Vovk, Ilia Nouretdinov, and Alex Gammerman. Testing exchangeability on-line. In *International Conference on Machine Learning*, 2003.

Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005. doi: 10.1007/b106715.

Ian Waudby-Smith and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society: Series B*, 86(1):1–27, 2024. doi: 10.1093/jrsssb/qkad009.

A Proof of Theorem 3

(a) Average run length. Each component detector $R_t^{(j)} = (1+R_{t-1}^{(j)})e_t^{(j)}$ satisfies $\mathbb{E}[R_t^{(j)} \mid \mathcal{F}_{t-1}] \le 1+R_{t-1}^{(j)}$ under the null, so the mixture $R_t = \sum_j \nu_j R_t^{(j)}$ does too, and $R_t - t$ is a supermartingale. Optional stopping at $T \wedge n$ gives $\mathbb{E}[R_{T\wedge n}] \le \mathbb{E}[T \wedge n]$. Since $R_{T\wedge n} \ge \frac{1}{\alpha}\mathbf{1}\{T \le n\}$, $\mathbb{P}(T \le n)/\alpha \le \mathbb{E}[T]$. Letting $n \to \infty$, either $\mathbb{P}(T < \infty) < 1$ and $\mathbb{E}_\infty[T] = \infty$, or $\mathbb{E}_\infty[T] \ge 1/\alpha$.

(b) Delay. Condition on $\mathcal{F}_\tau$. For every j, $R_{\tau+n}^{(j)} \ge \prod_{k=\tau+1}^{\tau+n} e_k^{(j)}$ because the recursion contains the term started at $\tau+1$ with weight one, so $R_{\tau+n} \ge \nu_j \prod_{k=\tau+1}^{\tau+n} e_k^{(j)}$ with $\nu_j = 1/m$.

Take $\lambda \le \lambda^\star = \min\{\Delta/(2v), 1/(2B')\}$ on the grid, so $\lambda|X_k| \le \frac{1}{2}$ and $\log(1+x) \ge x - x^2$ for $x \ge -\frac{1}{2}$ gives $\log e_k \ge Y_k := \lambda X_k - \lambda^2 X_k^2$. Then $\mathbb{E}[Y_k \mid \mathcal{F}_{k-1}] \ge \lambda\Delta - \lambda^2 v \ge \lambda\Delta/2$ since $\lambda v \le \Delta/2$. Let $D_k = Y_k - \mathbb{E}[Y_k \mid$

$\mathcal{F}_{k-1}]$. Because $|Y_k| \le \lambda B' + \lambda^2 B'^2 \le 1.5\lambda B'$, $|D_k| \le 3\lambda B'$. Because $Y_k^2 \le 2\lambda^2 X_k^2(1+\lambda^2 X_k^2) \le 2.5\lambda^2 X_k^2$, $\mathrm{Var}(D_k \mid \mathcal{F}_{k-1}) \le 2.5\lambda^2 v$. Freedman's inequality (Freedman, 1975) for the lower tail with $s = n\lambda\Delta/4$ gives

$$\mathbb{P}\Big(\sum_{k=\tau+1}^{\tau+n} D_k \le -s \;\Big|\; \mathcal{F}_\tau\Big) \le \exp\Big(-\frac{s^2}{2(2.5n\lambda^2 v + 3\lambda B' s/3)}\Big) = \exp\Big(-\frac{n\Delta^2}{80v + 8B'\Delta}\Big).$$

This is at most δ when $n \ge (80v + 8B'\Delta)\log(1/\delta)/\Delta^2$. On the complement, $\sum_k Y_k \ge n\lambda\Delta/2 - n\lambda\Delta/4 = n\lambda\Delta/4$.

The linear grid contains $\lambda_j = j/(mB') \in (\lambda^\star/2, \lambda^\star]$ with $j = \lfloor \lambda^\star m B' \rfloor \ge 1$, which holds because $\lambda^\star \ge 1/(mB')$ under $\Delta \ge 2v/(mB')$ and $m \ge 2$. With $\lambda = \lambda_j$, on the good event $\log R_{\tau+n} \ge n\lambda_j\Delta/4 - \log m \ge n\lambda^\star\Delta/8 - \log m$, which is at least $\log(1/\alpha)$ when $n \ge 8\log(m/\alpha)/(\lambda^\star\Delta)$. Hence $T \le \tau + n$.

(c) Horizon h updates once per h rounds on a chain; apply (b) to the chain with its margin and second moment, and add at most h rounds for the position of the changepoint within a chain step. □

B Experimental details

Simulator. Linear-Gaussian latent system $z_{t+1} = Az_t + Ba_t + w_t$, $w_t \sim \mathcal{N}(0, 0.1\, I_4)$, spectral radius of A equal to 0.9, exogenous exploration actions $a_t \sim \mathcal{N}(0, 0.3^2 I)$, fixed unit drift direction u. The model uses the pre-change parameters, so it is exactly faithful until an unmodelled force fu is added to the true update. The statistic is $Z = \frac{1}{d}(y-\mu)^\top \Sigma_h^{-1}(y-\mu) - 1$ clipped to $[-1, 6]$ with the exact h-step predictive covariance $\Sigma_h = \sum_{i<h} A^i Q (A^i)^\top$.

Closed-loop simulator. Same dynamics with state-dependent noise $q(z) = q_0(1 + \beta\|z\|^2/(\|z\|^2 + d))$, $q_0 = 0.1$, $\beta = 4$, a fully actuated $B = 0.8\, Q_{\mathrm{orth}}$, and certainty-equivalent one-step model-predictive control through the model (`atlas/closedloop.py`); actions are clipped to ± 2 with exploration noise 0.05.

Bets and thresholds. Mixture over $m = 20$ equally spaced λ on $(0, 1/(z_{\mathrm{hi}} - z_{\mathrm{lo}})]$; level $\alpha = 0.05$, per-horizon level α/H.

Reproduction. All simulator experiments run on CPU in minutes (`experiments/`, results in `results/`); the real-video replication needs the public Moving MNIST and KTH files.